Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking

Abstract

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**;
• **Computing methodologies** → **Rendering**; *Computer vision.*

Keywords

eye tracking, deep learning, computer graphics

ACM Reference Format:

. 2025. Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking. In *Proceedings of SIGGRAPH*. ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Problem

1.1 Challenges of reconstructing Purkinje reflections via 3D Gaussian Splatting (3DGS).

3DGS represents view-dependent appearance by using low-order Spherical Harmonics (SH) [Kerbl et al. 2023], therefore, accurately modeling high-frequency, view-dependent specular appearance remains challenging [Jiang et al. 2024; Tang et al. 2025]. More importantly, P3 and P4 are formed through transmission and reflection across the multiple ocular interfaces of eyes, whereas rasterization-based 3DGS does not explicitly model these secondary optical paths [Wu et al. 2025] (see Fig. X). In addition, Lu et al. [Lu et al. 2020] and Wu et al. [Cornsweet and Crane 1973] show that P3 is formed at a different optical focal plane from P1 and P4, while higher-order Purkinje reflections are relatively weak and therefore more susceptible to defocus and low contrast. Finally, Purkinje locations vary jointly with accommodation, vergence, subject-specific ocular geometry, and camera–illumination configuration [Lu et al. 2020; Ozhan et al. 2023]. Purkinje reflections are better characterized as dynamic optical signals governed by ocular optics rather than stable surface textures, making it difficult for standard 3DGS to reconstruct them consistently with their underlying physical behavior.

2 Introduction

3 Related works

We review works from two main areas: experimental approaches to ocular accommodation estimation and computational methods for virtual eye modeling and simulation. Table 1 compares reviewed works

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SIGGRAPH, 979-8-4007-1549-5/2025/08

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

3.1 Eye Accommodation Tracking

The depth-perception studies use eye-tracking and controlled optical setups to measure vergence responses under different perceived-depth conditions [Bontha and Arefin 2024; Sturgeon et al. 2025]. Vergence alone does not fully describe the accommodative state of the eye. Accommodation is harder to measure directly, especially in compact near-eye systems [Dunn 2019]. Purkinje images (P1-P2-P3-P4) provide a promising optical cue because they arise from reflections at ocular interfaces, including the crystalline lens surfaces, and vary with the lens geometry during accommodation. Lu et al. [2020] use a multi-camera, multi-LED AR-glasses prototype to track higher-order Purkinje images, extending conventional pupil–corneal-reflection tracking toward accommodation-aware sensing in a near-eye form factor. Ozhan et al. [2023] extend this direction by collecting multi-subject Purkinje measurements and combining a Purkinje imaging setup with a **Multi-Layer Perceptron (MLP)**-based regression model to estimate dynamic vergence and accommodation across users. Surveyed works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but they also reveal a practical challenge: *P3 and P4 are faint, easily defocused, and highly sensitive to LED–camera–eye geometry.* ToDo: Our work's differentiation goes here

3.2 Computational Eye Modeling

Computational eye models are widely used to generate labeled data for gaze estimation. UnityEyes, NVGaze, and RIT-Eyes synthesize controllable variations in gaze, ocular geometry, appearance, and illumination for training eye-tracking models [Kim et al. 2019; Nair et al. 2020; Wood et al. 2016]. These systems capture visible eye-region cues effectively, but they primarily target eye rotation, pupil localization, and segmentation rather than accommodation-dependent changes within the crystalline lens.

Neural eye representations improve visual realism and relighting. EyeNeRF combines an explicit eyeball surface with implicit volumetric representations for view synthesis, gaze animation, corneal refraction, and relighting [Li et al. 2022]. Khademi et al. extend a similar hybrid formulation to near-field point sources and structured illumination [Khademi et al. 2026]. However, these methods focus on photorealistic appearance and do not explicitly predict P3/P4 formation or accommodation-dependent internal ocular geometry.

Purkinje-based systems model internal reflections more directly. Wu et al., Idrizovi'c et al., and OpenIrisDPI use P1 and P4 for accurate eye-rotation tracking [Idrizović et al. 2025; Ressmeyer et al. 2026; Wu et al. 2023]. Lu et al. further model higher-order reflections with multiple cameras and infrared LEDs to estimate vergence and accommodation [Lu et al. 2020]. Ozhan et al. simulate P1–P4 across accommodation and vergence states, validate the predictions against real captures, and use the resulting features for dynamic estimation [Ozhan et al. 2023]. Our work follows this direction by

Table 1: Comparison of vergence (V) and accommodation (A) sensing approaches. Camera refers to imaging hardware per eye, from one camera to bench-top optics. Illum. refers to whether infrared sources are freely placed, fixed in a commercial tracker, or bench-mounted. Method refers to the sensing principle. Learning refers to whether the target is predicted by a learned model, a physical eye model, or a fixed formula. Accuracy refers to the reported error of the quantity in parentheses (high: $< 0.5^\circ$ gaze or < 0.25 D). Colours rate entries within each column for a compact wearable V/A sensor (green better, red worse).

Ref.	Camera	Illum.	Method	Data	Learning	V/A Target	Near-eye	Accuracy
[Cornsweet and Crane 1973]	Bench	Bench source	Purkinje	Real	Physical model	Neither	No	High (gaze)
[Crane and Steele 1978]	Bench	Bench source	Purkinje	Real	Physical model	A only	No	High (A)
[Schaeffel et al. 1987]	Bench	Bench source	Photorefraction	Real	Physical model	Both	No	Mid (A)
[Morimoto et al. 2002]	1/eye	Movable LEDs	Pupil/glint	Real	Physical model	Neither	No	Mid (gaze)
[Rosales et al. 2006]	Bench	Bench source	Purkinje	Real	Physical model	A only	No	Mid (A)
[Świrski and Dodgson 2013]	1/eye	Fixed LEDs	Pupil/glint	Real	Physical model	Neither	Yes	Mid (gaze)
[Weier et al. 2018]	1/eye	Fixed LEDs	Pupil/glint	Real	Regression	V only	Yes	Mid (V)
[Kim et al. 2019]	1/eye	Movable LEDs	Pupil/glint	Both	CNN	Neither	Yes	Mid (gaze)
[Padmanaban et al. 2019]	1/eye	Fixed LEDs	Pupil/glint	Real	Formula	V only	Yes	Mid (V)
[Lu et al. 2020]	≥2/eye	Movable LEDs	Purkinje	Synthetic	NN regression	Both	Yes	Mid (A)
[Angelopoulos et al. 2021]	≥2/eye	Movable LEDs	Pupil/glint	Real	Physical model	Neither	Yes	High (gaze)
[Wu et al. 2023]	Bench	Bench source	Purkinje	Real	Physical model	Neither	No	High (gaze)
[Ozhan et al. 2023]	1/eye	Movable LEDs	Purkinje	Real	ML regression	Both	No	High (A)
[Ahn and Lee 2025]	1/eye	Movable LEDs	Purkinje	Real	Regression	V only	No	Low (V)
[Sturgeon et al. 2025]	1/eye	Fixed LEDs	Pupil/glint	Real	Formula	V only	Yes	Low (V)
[Ressmeyer et al. 2026]	Bench	Bench source	Purkinje	Real	Physical model	Neither	No	High (gaze)
Ours	1/eye	Movable LEDs	Purkinje	Both	3DGS	Both	Yes	High (A)

Table 2: Comparison of eye models. Representation refers to how far the model is fitted to an individual eye: geometry and appearance from images, appearance on a template eyeball, or population parameters only. Purkinje refers to which reflections the model reproduces; only lens reflections (P3, P4) carry accommodation information. Input refers to the images needed to personalise the model. Fidelity refers to agreement with real images. Speed refers to rendering rate (real-time: >30 fps). Gaze refers to whether eye rotation is a controllable variable. Colours rate entries within each column for an eye model fitted to a real eye that reproduces lens reflections under controlled gaze (green better, red worse).

Ref.	Method	Representation	Purkinje	Input	Fidelity	Speed	Gaze
[Navarro et al. 1985]	Navarro schematic eye	Optical model	P1–P4	Synthetic	Low	Interactive	Yes
[Wood et al. 2016]	UnityEyes	Parametric mesh	P1	Synthetic	Low	Real-time	Yes
[Nair et al. 2020]	RIT-Eyes	Parametric mesh	P1	Synthetic	Mid	Offline	Yes
[Sharifzadeh 2024]	Purkinje ray model	Optical model	P1	Synthetic	Low	Interactive	No
[Li et al. 2022]	EyeNeRF	Mesh + NeRF	P1	Light stage	High	Offline	Yes
[Ruzzi et al. 2023]	GazeNeRF	NeRF	None	Few images	Mid	Offline	Yes
[Lin et al. 2026]	NeRF–mesh eye simulator	Mesh + NeRF	P1	Light stage	Mid	Offline	Partial
[Khademi et al. 2026]	Relightable hybrid eye	Mesh + NeRF	P1	Light stage	High	Offline	No
[Wang et al. 2024]	GazeGaussian	Gaussians	None	Few images	Mid	Real-time	Yes
Ours	Purkinje 3DGS	Gaussians + deform.	P1–P4	Multi-view	High	Real-time	Yes

using a controllable optical eye model to predict how accommodation, vergence, and LED–camera geometry affect P1–P4 formation.

3.3 Physics-Based Optical Design

Differentiable optical design provides a possible framework for optimizing the illumination and imaging layout of a Purkinje capture system. Na et al. show that image-domain losses can be propagated through differentiable ray tracing to physical optical parameters [Na et al. 2024]. For our system, the same principle could be applied to LED position, camera pose, focal distance, aperture, and field of view.

The optimization criteria should follow the capture behavior of higher-order Purkinje reflections. Lu et al. show that P3 is weaker than P1, lies on a different focal plane, and may leave the camera field of view as gaze changes; their system therefore uses multiple cameras and infrared LEDs [Lu et al. 2020]. Ozhan et al. further show that the camera–LED–eye geometry affects the observed P3/P4 locations and configure the system to improve reflection separation while avoiding vignetting [Ozhan et al. 2023]. These results suggest optimizing P3/P4 visibility, sharpness, field-of-view coverage, and separation from other reflections.

The selected layout must also remain practical to assemble, calibrate, and evaluate at the sensor level. Prior work on replicating a controllable AR haploscope highlights the implementation challenges of physical optical platforms [Bontha and Arefin 2024] and

Wu et al. show that reliable P1/P4 measurement depends on sufficient image resolution and accurate reflection localization [Wu et al. 2023]. These considerations motivate evaluating optical layout, calibration, and sensor-level detectability together.

The present work does not perform end-to-end differentiable optimization. Instead, it uses geometric-optics simulation to compare candidate LED–camera configurations and to predict whether P1–P4 remain visible, sharp, and separable under realizable capture conditions. Differentiable layout optimization and synthetic-to-real adaptation are left as future extensions.

4 Optical Background

4.1 Capturing Purkinje image

Dual Purkinje Image (DPI) tracking can provide higher precision than conventional pupil–corneal-reflection tracking by using the relative motion of P1 and P4 rather than the pupil center [Cornsweet and Crane 1973; Ressmeyer et al. 2026]. However, obtaining reliable Purkinje features from real eye images remains challenging. Wu et al. [Cornsweet and Crane 1973] show that P1 and P4 can be imaged together while P3 is out of focus and P2 largely overlaps with P1; similarly, Ahn et al. [Ahn and Lee 2025] report that P4 is substantially smaller and darker than P1, making its detection less reliable. Therefore, the existence of a valid reflection path does not necessarily guarantee a usable image feature, as its visibility further depends on optical focus, reflection strength, occlusion, and imaging noise [Ahn and Lee 2025; Cornsweet and Crane 1973]. Camera and illumination configuration also affect whether these reflections remain observable across different eye orientations, making Purkinje detectability a joint consequence of ocular geometry and the imaging configuration [Cornsweet and Crane 1973; Lu et al. 2020]. Nevertheless, Purkinje geometry carries useful information about the underlying ocular surfaces, with previous phakometric studies showing that the position and geometry of these reflections can be used to infer ocular surface curvature and alignment [Barry et al. 2001].

4.2 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since P1–P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periocular appearance modeling [Khademi et al. 2026; Li et al. 2022; Nair et al. 2020]. Existing Purkinje imaging systems also provide practical guidance for hardware geometry. Both Lu et al. and Ozhan et al. place infrared LEDs close to the camera optical axis to maximize Purkinje visibility while keeping the reflections inside the camera field of view. At the same time, camera positions must balance the visibility of higher-order Purkinje images against eyelid and eyelash occlusions, glint overlap, and sensor saturation. These observations suggest that LED and camera placement is not arbitrary but jointly determines the brightness, separation, and detectability of P1–P4. Once the capture geometry is determined, synthetic rendering provides an efficient way to evaluate different configurations before physical experiments. RIT-Eyes demonstrates that physically based rendering can generate realistic near-eye images under controllable illumination, while EyeNeRF and subsequent hybrid eye representations further show that explicit corneal geometry is necessary for reproducing specular reflections and refraction, whereas implicit neural representations are better suited for modeling complex periocular appearance and relighting. These observations motivate the use of explicit optical eye models for Purkinje simulation before subsequent NeRF or 3D Gaussian reconstruction, fall outside the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023]

5 Method

We adopt 3D Gaussian Splatting (3DGS) as the base representation for reconstructing the eye scene [Kerbl et al. 2023]. The scene is represented by a set of anisotropic 3D Gaussians, each parameterized by its position $\boldsymbol{\mu}$, covariance Σ defined by scale and rotation, opacity α, and spherical harmonics (SH) coefficients for view-dependent appearance [Jiang et al. 2024; Kerbl et al. 2023].

During rendering, the 3D Gaussians are projected onto the image plane, depth-sorted, and accumulated using front-to-back α-compositing [Kerbl et al. 2023]:

$$\mathbf{C}(\mathbf{p}) = \sum_i T_i(\mathbf{p})\alpha_i(\mathbf{p})\mathbf{c}_i(\mathbf{v}),$$

where $\mathbf{c}_i(\mathbf{v})$ denotes the view-dependent color predicted from the SH coefficients. Training jointly optimizes Gaussian positions, scales, rotations, opacities, and SH coefficients using a combination of L_1 and D-SSIM reconstruction losses, while adaptive densification increases the number of Gaussians in regions requiring higher representational capacity [Kerbl et al. 2023]. We use this standard 3DGS formulation as our baseline and subsequently introduce Purkinje-specific local supervision.

5.1 3D Gaussian Splatting

3D Gaussian Splatting is an explicit 3D scene representation and differentiable rendering method designed for novel view synthesis. The core idea is modeling the geometric structure and viewpoint-dependent appearance of a scene in 3D space using a set of optimizable Gaussian primitives expressed as

$$\mathcal{G}_i = \left\{\boldsymbol{\mu}_i,\ \Sigma_i,\ \alpha_i,\ \mathbf{c}_i\right\}, \tag{1}$$

Here, $\boldsymbol{\mu}_i$ denotes the 3D position of the i-th Gaussian, $\mathbf{s}_i$ its scale, $\mathbf{R}_i$ its rotation, α_i its opacity, and $\mathbf{c}_i$ the spherical harmonics coefficients used to represent view-dependent appearance.

The input to 3DGS consists of a set of multi-view images capturing the same scene from different viewpoints, structure-from-Motion (SfM) estimates the camera parameters and reconstructs a

sparse 3D point cloud, which is subsequently used to initialize the Gaussian primitives. [Jiang et al. 2024; Kerbl et al. 2023]. The scale and rotation jointly define the covariance matrix as

$$\Sigma_i = \mathbf{R}_i \mathbf{S}_i \mathbf{S}_i^T \mathbf{R}_i^T, \tag{2}$$

where $\mathbf{S}_i$ is a diagonal scaling matrix constructed from $\mathbf{s}_i$, and $\mathbf{R}_i$ determines the orientation of the anisotropic Gaussian.

Once the Gaussian primitives are initialized and parameterized, they are projected onto the image plane, ordered according to depth, and accumulated using front-to-back alpha compositing to generate the rendered image. Standard 3DGS combines a pixel-wise L_1 loss with a D-SSIM loss [Kerbl et al. 2023]:

$$\mathcal{L}_{\text{GS}} = (1-\lambda)\mathcal{L}_1 + \lambda \mathcal{L}_{\text{D-SSIM}}, \tag{3}$$

where $\mathcal{L}_1$ measures the pixel-wise photometric difference between the rendered and ground-truth images, $\mathcal{L}_{\text{D-SSIM}}$ measures their local structural dissimilarity, and λ controls the relative contribution of the two terms. The reconstruction loss is back-propagated to jointly optimize the Gaussian positions, scales, rotations, opacities, and SH coefficients.

Before the formal training, we conducted ablation experiments to compare four 3DGS models. Using Splatfacto from Nerfstudio as the baseline, we first used the default configuration and then enabled all the options to test whether a single Gaussian set could simultaneously represent iris texture and Purkinje images when the capacity was sufficient. Subsequently, we compared two variants that separately modeled reflection and transmission: RT-Splatting divided the Gaussians into a reflection layer and a transmission layer, rendered them separately and then composited them; TransparentGS explicitly handled reflection and refraction for transparent objects. The evaluation indicators were the position errors of P1 and P4 at the validation views, as well as the difference in PSNR between the training views and the validation views.

5.2 The collection of synthetic eye data and real eye images

We built a Blender model with LuxCore path tracing to parameterize the eye model, including the four refractive interfaces (the anterior and posterior surfaces of the cornea, and the anterior and posterior surfaces of the lens), whose parameters were selected based on the Arizona eye model.

Besides, the gaze is controlled by the rotation of the eyeball around the rotational center, and the accommodation is controlled by the radius of curvature and thickness of the front and back surfaces of the lens, which change with the diopter value. Other than that, in order to create various eye states, we set the pupil size of normal people under extreme conditions to 2 mm, and at the same time modify four parameters including the position of the LED and camera, the rotation angle and the accommodation. We experimented with different accommodation and convergence methods to actively set the eyes under the Vergence-Accommodation Conflict (VAC) condition to obtain the Purkinje images. However, Blender is not verified optical software. Therefore, the same set of eye model parameters was created using Zemax OpticStudio for ray tracing.

Then, the positions and sizes of P1–P4 given by both methods were compared; the model is accepted only if the deviation within the sampling range is less than one pixel. On the other hand, the binocular experimental paradigm used in the collection is as follows: The subjects are fixed on a chin rest, and a near-infrared camera is used in conjunction with a fixed point light source for filming; the gaze target is presented at different distances as physical objects within the range of 0.3–4 meters, corresponding to a range of 0.25–3 D. Each distance is held for several seconds, and a motor-driven Thorlabs device rotates the camera at a position 5 cm from the eye, with a scanning range of ±30 degrees centered on the eye's center. Then, different individuals are selected to repeat the experiment to obtain diverse eye data.

5.3 CNN dynamic contrast enhancement

The P3 and P4 values of the real image are approximately two orders of magnitude darker than P1, and would be ignored in direct training. First, we use Pupil Labs to detect the pupils and the iris area, and then apply a lightweight CNN to perform dynamic contrast enhancement on this area: the network takes the original frame and the pupil mask as input, and outputs the local gain and bias for each pixel, so that the dim reflections within the pupil are lifted, the saturated area of P1 is reduced, and the training objective is to make the enhanced image consistent with the clean rendering on the synthetic data. Then the preprocessed images are used to train the 3DGS model to generate synthetic eye data.

5.4 Purkinje image reconstruction: From synthetic domain to real domain

Referencing NVGaze, we first train on synthetic data, then adapt with a small amount of real data. In the first stage, we train the 3DGS using large-scale synthetic images (with possible degradation including lens aberration, motion blur, and Gaussian readout noise) with physical restrictions (the tissue layer rotates with the rigid body of the eyeball, and the position of the reflective layer is bound to the centers of curvature of the cornea and lens and moves with gaze and light source). In the second stage, we add the enhanced real images from Section 3.3 to the training, sharing the same loss, only allowing the anchor points of the reflective layer to be fine-tuned within a small range, and freezing the residual network learned in the synthetic stage, and finally generate an effective 3DGS model.

6 Results and Discussions

7 Results and Discussions

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